

Recitation 11 — Answer Key

1): Part I Volcker's Disinflation

Part a)

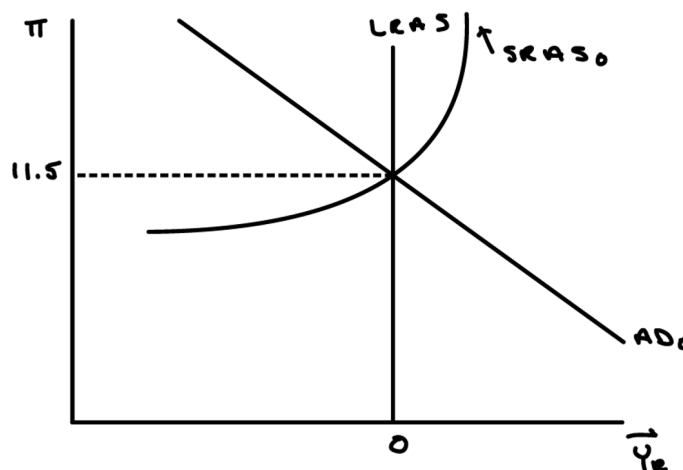
Model the U.S. economy in the late-1970s where actual inflation and expected inflation are both 11.5%, and real GDP growth is 0%.

Answer:

In the late-1970s:

- Inflation expectations (π^e) are anchored at 11.5% due to years of high inflation
- Actual inflation (π) is also 11.5% — expectations match reality
- Real GDP growth is 0% — the economy is at its potential output
- The economy is in long-run equilibrium with high inflation
- The AD and SRAS curves intersect at the LRAS

Graph:



Part b)

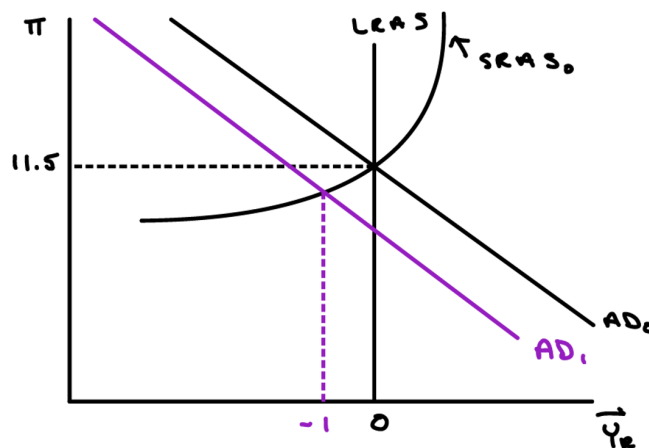
In 1979, Paul Volcker was appointed chair of the Federal Reserve Board. What impact do you think Volcker's policy had on real economic growth? Show the impact on the graph from part a.

Answer:

Volcker implemented strong **contractionary monetary policy** by sharply reducing the money supply growth rate.

- This is a left-ward shift of the AD curve from AD_0 to AD_1
- The economy moves to a point where inflation remains elevated but real growth falls sharply
- In the short run, the economy experiences a **recession**
- Real GDP growth becomes negative (the economy contracts)
- This was a deliberate, painful policy choice to break the back of high inflation

Key Insight: There is a short-run trade-off between inflation and growth. Fighting inflation requires accepting lower growth/higher unemployment in the short run.



Part c)

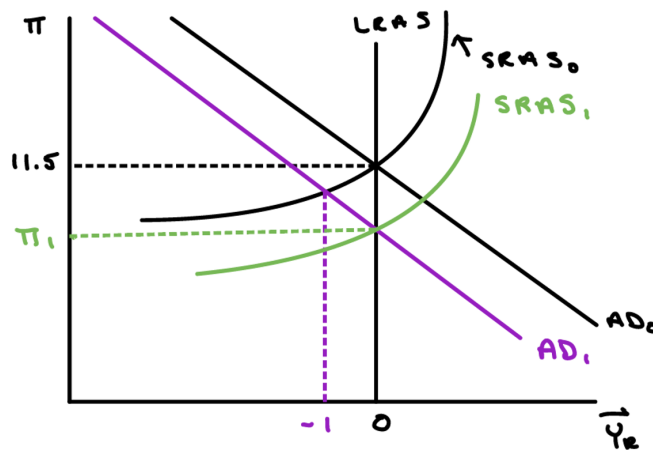
What impact do you think Volcker's monetary policy had on inflation expectations and actual inflation in the early-1980s? Use the graph from parts a and b to illustrate your answer.

Answer:

Over time, Volcker's sustained contractionary policy succeeded in reducing inflation expectations:

1. As inflation remains lower for several periods, households and firms gradually revise expectations downward
2. Workers accept lower wage growth; firms moderate price increases
3. The SRAS curve shifts **downward** from $SRAS_0$ to $SRAS_1$
4. Actual inflation falls to π_1 (a lower level)
5. Real growth gradually recovers toward potential as inflation expectations fall
6. The economy transitions from the recession toward a new long-run equilibrium at lower inflation

Final Graph (Long-run impact):



Summary: Volcker's policy demonstrates how disinflation works in practice. In the short run, fighting inflation is costly in terms of lost growth and jobs (the recession of 1981-82).

However, by maintaining the restrictive stance long enough to change inflation expectations, the economy eventually stabilizes at a lower inflation rate with restored growth. This highlights the importance of central bank credibility in anchoring expectations.

2): Part II

Q1: Correct Answer: Leftward shift of AD ; Output falls due to “sticky” prices.

Rationale: The loss of trillions in wealth reduces consumption (C), shifting AD left. Because wages and prices are sticky and far from perfectly flexible, the economy moves down along the upward-sloping AS curve into a recessionary gap rather than immediately adjusting prices downward[cite: 134, 161].

Q2: Correct Answer: Leftward shift of AD .

Rationale: A credit freeze or disruption in the “cost of credit intermediation” causes Investment (I) to plummet. Because businesses and consumers cannot access the global “Giant Pool of Money,” total spending drops, shifting AD to the left.

Q3: Correct Answer: Prices would drop, but output would remain the same.

Rationale: In a Classical model with a vertical AS curve, a drop in AD results only in immediate deflation. Real output (Y) would remain at the potential level because, in this theoretical case, there are no nominal rigidities to prevent instantaneous price adjustment.

Q4: Correct Answer: Rightward shift of AD ; Increase real output (Y).

Rationale: Expansionary monetary policy increases the money supply, which lowers real interest rates. This stimulates I and C , pushing the AD curve back to the right to restore equilibrium at potential output.